EART11300 Practice Practical Exam Questions on Rocks & Minerals

Minerals

Models 3 and 10 are models of two different minerals, showing their ideal shape. What is the most appropriate habit term for each of them?

Match the shapes with the closest ones on page 1 of the minerals practical handout.

Determine the hardness of mineral M2. Give your answers as a number on Mohs' hardness scale, to the nearest whole number, and explain your answer. A copper coin and glass slide are available to help you if you need them.

Make sure you test both ways, e.g. if you rub the crystal across the glass slide, it might look like you are scratching it when in fact you are leaving a deposit from the crystal on the slide, so check whether you can get a corner of the slide to scratch the mineral. You might need to press quite hard, and in a controlled way, i.e. try to dig a little trench rather than aimlessly pushing the material around.

Describe the cleavage of mineral M9:

Remember to indicate how good the cleavage is. You could compare with your own mineral samples. Also note the clear distinction between "at 90° " and "not at 90° " – in the case of the latter, you wouldn't need to specifiy the actual angle, just that it is not 90°.

Rocks

For each of the hand specimens provided, indicate what kind of rock it (igneous, sedimentary or metamorphic) and explain your answer.

Your reasons should leave me in no doubt that it must be whichever rock type you say it is. For example, some characteristics apply to >1 rock type, so are not enough on their own. Also, you cannot say "I know it is sedimentary because it is chalk". You need to give the properties (Is it made of clasts? Is it porous? etc.) that allowed you to say that it is sedimentary.

Draw an annotated sketch of ϕ17 or ϕ18. Your sketch should show one or more of the textural features that enable you to determine whether the rock is igneous, sedimentary or metamorphic.

Remember a scale on your sketch. I usually wait until I have finished my sketch, and then pick something on the sketch that I can measure, and put a scale bar beside it (e.g. see sketch below). Some people like to draw a box before they start – it is entirely up to you what you do. Remember, it doesn't need to be perfect (it is not a photo) – it is intended to highlight the important features. Try to make sure that everything on your sketch is significant, e.g. you don't need to draw the edges of the rock; if there is something weird-looking that you cannot identify, then maybe leave it out. And unless you are a really good artist, then if you do not need to show something in 3-d, just draw a 2-d surface.

